

YUSEF SYED

Toronto, ON | U.S.-Canadian dual citizen

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Education

University of Toronto

Expected May 2030

Mathematics and Computer Science (Intended Double Major)

Faculty of Arts & Science, first-year Mathematical & Physical Sciences

Current coursework: programming, calculus with proofs, linear algebra, statistical reasoning. Prior coursework: Data Structures, Lehigh Carbon Community College (dual enrollment).

Selected Research and Engineering Projects

Agent Eval Mutation Lab | [Code and evidence](#)

Aug 2026 - Present

Python, Inspect, Docker, SQLite

- Built a resumable Python evaluation harness separating tool proposals, observed effects and unknown outcomes; tested **13 hand-authored synthetic cases** under four evidence conditions.
- Integrated Inspect with Docker-isolated tools and independent SQLite checks to detect state changes that occurred before an error or were later revoked.
- Published a **624-trial local-model study** with a pre-specified, version-controlled protocol; withheld claims of improvement because invalid-output rates differed between prompting conditions.

Tiraz Garment Completion Study | [Code and results](#)

Aug 2026

Python, PyTorch, NumPy

- Trained three Deep Sets-style models on **27,683 source groups from Fashionpedia annotations**; achieved **58.5-59.0%** top-1 on 962 held-out groups versus a **52.7%** co-occurrence baseline.
- Separated training, tuning and calibration groups. At a 90% coverage target, prediction-set coverage fell from **91.1-91.2% to 55.6-57.9%** after withholding additional visible categories.

ShiftProof | [Code and evaluation](#)

Sep 2026

Python, local LLM, constraint verification

- Built a local LLM scheduling assistant with independent constraint checks and coordinator approval before changes; excluded untrusted names and notes from model context.

Open-Source Contributions

Contributor | [Merged contributions](#)

Aug 2026 - Present

- Contributed **9 merged upstream PRs** across Microsoft Agent Lightning, Microsoft PyRIT, W&B RAI Toolkit, Meridian Labs Inspect Scout and Kornia.
- Fixed asynchronous child scoring and unassessed-result handling in W&B's [CompositeScorer](#); added focused regressions for aggregation and async behavior.
- **Open PyTorch PR:** added CPU/CUDA shape gradients for regularized incomplete gamma. Author-run source-build tests checked two derivatives on **581 inputs** across macOS CPU, Linux CPU and RTX A4000 CUDA against reference values. [PR](#) | [Validation](#)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL

Research and testing: PyTorch, NumPy, Inspect, pytest, mypy

Development tools: Codex, Git, GitHub Actions

Systems: Docker, SQLite, PostgreSQL

Selected Leadership

Feeding the Valley | Vice President

2025-2026

Supported fundraising and meal preparation for local food banks.

FIRST Robotics Team 834 | Fundraising Lead

2022-2026

Coordinated sponsorship outreach and crowdfunding.